

EXCEL ARCHITECTURE WORKSHOP

MASTERING DATA LOGIC + AI LEVERAGE | MICHAEL SMITH, M.S. ED.

Transform messy spreadsheets into automated data engines. This curriculum focuses on building robust logical foundations and leveraging AI to move from data entry to strategic analysis.

PHASE 1: THE CORE FOUNDATION

1.1 STRUCTURAL INTEGRITY

Stop using Excel as a typewriter. Master official Tables (Ctrl+T) to create "Self-Healing" data ranges that expand and update automatically.

1.2 LOGIC FRAMEWORK

Master Absolute References (\$) and Data Validation. Build "Guardrails" that prevent user errors and ensure your data plumbing is leak-proof.

PHASE 2: AI LEVERAGE & MANAGEMENT

2.1 THE DATA ANALYST

Use AI to architect complex formulas and scripts. Move beyond memorizing syntax and learn to describe logical outcomes in plain English.

2.2 THE VISIONARY

Leverage AI to identify trends, outliers, and hidden stories. Use the "Verification Protocol" to ensure machine insights are grounded in reality.

Phase 3: The Custom Master Build

The Educator Build

Automated gradebooks with "At-Risk" alarm systems. AI-generated progress comments and longitudinal data tracking.

The Student Build

Stress-testing financial models and scientific data. Using "What-If" scenarios to predict future project outcomes.

The Professional Build

One-page Command Centers (Dashboards). Master Pivot Tables and Slicers to deliver answers, not just raw rows.